Terrain-interaction control of a hexapod robot: a measured comparison of three controllers on a shared simulation plant

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Abstract

This work presents a comparative analysis of three control architectures of increasing complexity for the PhantomX AX Metal Mark II hexapod, developed and evaluated in a high-fidelity Simscape Multibody simulation environment. The modular progression comprises an open-loop kinematic tripod controller (C1), a torque-based terrain-adaptation feedback (C2) [1] and an upper attitude-control loop (C3), making it possible to isolate experimentally the contribution of each mechanism.

A fundamental part of the work concerned the identification and correction of two non-obvious modelling flaws of the test bench, capable of significantly altering the performance metrics. With a protocol that separates real differences from simulation variability, the experimental results highlight the precise trade-offs of the three approaches: terrain adaptation improves efficiency on discontinuities at the expense of attitude stability on flat surfaces, while adding the attitude loop restores body orientation at the price of higher energy consumption. Finally, the structural limits of all the controllers in recovering from lateral impulses are discussed.

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1 Introduction

1.1 The platform and the reference work

The PhantomX AX Metal Hexapod Mark II is an open-hardware hexapod robot with eighteen Dynamixel AX-12A servomotors, three per leg (coxa, femur, tibia) [3]. In its standard configuration it has no force sensors at the feet and no IMU: the only feedback available from the hardware is the position and the load estimate reported by each servo.

The reference paper [1] exploits exactly this constraint. Its contribution is a feedback architecture that detects foot–ground contact *from the motor torque alone*, comparing the torque measured at the joints with the one predicted by an inverse-dynamics model of the robot, and uses the detection to adapt the commanded foot height to a terrain that the trajectory generator does not know.

1.2 Why we chose this project

Three main reasons:

The paper does not highlight the cost of the mechanism.
The reference work shows that terrain search improves the behaviour on unmodelled terrain. It does not report how much the mechanism costs where there is no terrain to discover. A controlled comparison on identical terrain, with a declared validity criterion, can answer — and the answer turned out to be a real and systematic cost (Section 5).

The contact detector is a model-based observer.
The flag that drives the whole architecture is $\mathrm{OR}_j \left(|\tau_j^{\mathrm{meas}} - \tau_j^{\mathrm{exp}}| > \sigma_j \right)$, where τ^{exp} comes from an inverse-dynamics model with a rigid-body structure. Its quality therefore depends on the *accuracy of a dynamic model* (cf. Section 4.2).

A hexapod on a shared plant makes the comparison clean.
Since all three controllers drive the same multibody model, the same terrain, the same contact law and the same gait generator, every comparison between adjacent controllers differs by a single mechanism.

1.3 What this report claims

1. Two flaws of the simulation bench, found, measured, corrected and documented, each of which had already produced publishable-looking but wrong numbers (Section 4).
2. A measurement protocol that makes a difference between two controllers *declarable or not declarable*, rather than merely observed (Section 3).
3. A three-way comparison on seven tasks with that protocol applied (Section 5).
4. A feasibility check of the commanded torques against the datasheet of the real servo (Section 6).
5. A further test of the controller's potential by simulating a load-transport scenario (Section 7).
6. A stability analysis with ZMP evaluation (Section 8).

2 Experimental setup

2.1 The plant

The robot is imported into Simscape Multibody [4] from the public URDF of the PhantomX description package: 25 rigid bodies (trunk + 24 leg links), 18 revolute joints and six spherical contact elements at the feet. The joints are *position*-commanded; the resulting joint torque is

therefore a reaction, not an input. The foot–ground interaction uses a penalty contact law with Coulomb friction, tuned so that the static penetration (4.3 mm in tripod stance) lies well above the transition region (2.0 mm), where the force law would otherwise be ill-defined.

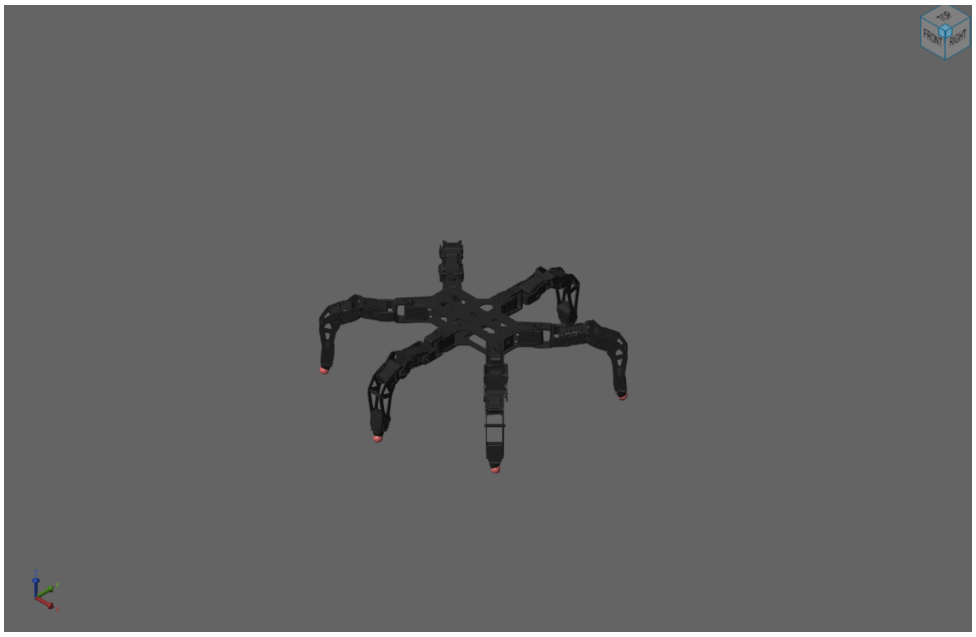


Figure 1: The robot as seen in Simscape Multibody.

2.2 The three controllers

Table 1: The three controllers. Each one adds a mechanism to the previous one.

	Plant	Mechanism
C1	`sim_zero`	Open-loop kinematic tripod. Foot trajectories generated from the gait parameters, per-leg inverse kinematics, position-commanded joints. No feedback of any kind.
C2	`sim_zero`	C1 + *terrain search*: when the foot has reached the commanded stance height but the flag is still low, the commanded foot height keeps descending until the torque residual exceeds the threshold.
C3	`sim_attitude`	C2 + an *attitude loop*: two discrete PI regulators on body pitch and roll, whose outputs are distributed over the six feet as an additive height offset.

For controller C3, the z we impose on each foot is given by the sum of two contributions: one coming from the terrain-search controller and the other from the attitude PI controllers.

$$\texttt{ricerca_terreno} \rightarrow z_i \rightarrow \Sigma(+\Delta z_i) \rightarrow \texttt{Saturation} \rightarrow \texttt{Rate Limiter} \rightarrow \text{leg}$$

with

$$\Delta z = \mathbf{x}\, u_\theta + \mathbf{y}\, u_\phi, \tag{1}$$

where u_θ and u_ϕ are the outputs of the pitch and roll regulators and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^6$ are the lever arms of the six feet. Pitch uses $P = 10$, $I = 1$; roll $P = 5$, $I = 0.5$; both are discrete at 1 ms with zero reference. Since C3 contains C2, the comparison C2 $\rightarrow$ C3 isolates the attitude loop and C1 $\rightarrow$ C2 isolates terrain search.

2.3 Gait parameters and their origin

The reference work delegates gait generation to NUKE and states explicitly that it will not discuss it. The gait parameters therefore had to be chosen. They are not arbitrary: three out of five are fixed by a measured constraint, and the remaining degree of freedom is only one, because

$$T = \frac{S}{\beta_{\mathrm{st}}\, v_{\mathrm{nom}}}. \tag{2}$$

Table 2: Gait parameters and their origin.

Parameter	Value	Origin
phase vector	—	Definition of the alternating tripod: groups $\{\mathrm{LF}, \mathrm{RM}, \mathrm{LR}\}$ and $\{\mathrm{RF}, \mathrm{LM}, \mathrm{RR}\}$. It is not a free parameter.
β_{st}	0.50	The only duty factor that gives continuous support on *exactly* three legs. Below it, there are instants with fewer than three feet on the ground (outside static stability); above it, six-foot windows appear in which 18 position constraints act on 6 degrees of freedom.
S	0.06 m	Limited by leg extension. At mid-step the leg uses 80.9% of $\ell_f + \ell_t = 0.219\,\mathrm{m}$; the internal assert fires at 85%, i.e. at $S = 0.095\,\mathrm{m}$. The 85% limit is itself measured: at 89% the robot walked backwards.
H	0.05 m	1.5× the measured obstacle height.
T	1.00 s	Dependent variable, through (2), given $v_{\mathrm{nom}} = 0.12\,\mathrm{m/s}$.

The resulting Froude number is $v^2/(gh) = 0.12^2/(9.81 \times 0.154) = 0.0095$: the inertial forces are one percent of gravity. The gait is clearly quasi-static, which is what legitimises the *static* stability assumption on which the tripod gait is built.

2.4 List of tasks

Table 3: The seven tasks.

ID	Name	What it stresses
T2	flat, speed sweep	Reference, and stability envelope in speed.
T3	curved path	Yaw command 0.1 rad/s.
T4	8° ramp	Continuous slope: body attitude versus terrain.
T4D	8°/8° bump	Ascent and then descent, two angular interfaces.
T5	single obstacle	An unmodelled step, 32.8 mm.
T6	seven obstacles	Four full-width, three lateral
T7	lateral impulse	Momentum step from 1% to 1600% of nominal.

3 Validation protocol and variability band

To ensure that the performance differences observed between the controllers are attributable to the control architecture and not to artefacts or fortuitous synchronies of the Multibody simulation, a strict validation protocol was defined, based on the computation of the *variability band*.

3.1 Motivation of the protocol

In a high-fidelity simulation environment, testing different controllers on mathematically identical terrain and at exactly the same height can lead to misleading conclusions. A specific combination of the foot impact instant, the solver discretisation and the micro-topography of the surface could artificially favour one algorithm over another.

Introducing a micro-perturbation on the nominal stance depth makes it possible to perform a sensitivity stress test, checking whether a controller's advantage remains solid even in the face of small unpredictable geometric variations.

3.2 Computation of the variability band

For each controller, on four tasks, the simulation is run three times keeping the control law unchanged and varying only the nominal stance depth z_0:

$$z_0 \in \{z_{\text{nom}} - 0.2\,\text{mm},\ z_{\text{nom}},\ z_{\text{nom}} + 0.2\,\text{mm}\} \tag{3}$$

This variation of $\pm 0.2\,\text{mm}$ does not alter the nature of the path, but changes the precise instant and height of the feet's contact with the ground.

The **variability band** σ_{var} associated with a generic metric M (for example the maximum pitch or the consumed energy) is computed as the maximum (peak-to-peak) excursion measured across the three runs of the same controller:

$$\sigma_{\text{var}}(M) = \max_{z_0}(M) - \min_{z_0}(M) \tag{4}$$

This value quantifies the intrinsic uncertainty of the measurement and defines the natural oscillation threshold of the metric due solely to a small variation of the contact conditions.

3.3 Validity criterion

To distinguish a real algorithmic improvement from mere simulation variability, a strict validation rule is applied: the difference between the performance of two controllers A and B on metric M is declared *confirmed and valid* only if it exceeds by at least three times the worse of the two recorded variability bands:

$$|M_A - M_B| \geq 3 \cdot \max\left(\sigma_{\text{var},A}(M),\ \sigma_{\text{var},B}(M)\right) \tag{5}$$

Should the gap be below this threshold ($< 3\times$), the variation is classified as *not conclusive* (n.c.), since it is not sufficiently larger than the variability band of the measurement and does not allow a definite merit to be attributed to the control architecture.

4 Corrections with respect to the original material

4.1 Correction of the URDF inertias

The plant takes masses and inertias from a public URDF. The masses are correct; the inertia tensors are not.

Radius of gyration.
From $I = mr^2$, the declared trunk inertia places the trunk mass at $r = 1.79\,\mathrm{m}$ from its own centre. The trunk is 25 cm wide. Each link gives $r = 0.46\,\mathrm{m}$; the links are 2–12 cm long. Read in $\mathrm{kg}\cdot\mathrm{m}^2$, as the URDF standard requires, those numbers describe an object that cannot exist.

All 24 links carry identical inertia,
to the same decimal. Coxa, femur and tibia differ in shape and length. It is a placeholder, not a measurement.

Dividing by 10^3 brings everything back into the physically plausible range, which is the signature of a CAD export performed with masses in grams. The effect on the conclusions, and not only on the values, is large: with the URDF inertias the pitch of C1 on rough terrain reached 45°, versus 15° with the corrected ones, and the cost ratio between C2 and C1 went from 1.35 to 0.68.

4.2 Correction of the contact detector

The contact flag of C2 is not a threshold on the measured torque. The model computes

$$\text{contact} = \bigvee_{j\in\{c,f,t\}} \left(\left| \tau_j^{\text{meas}} - \tau_j^{\text{exp}} \right| > \sigma_j \right), \tag{6}$$

and τ^{exp} is produced by an inverse-dynamics block that carries a `rigidBodyTree` imported *from the same URDF*.

We had corrected the inertias of the simulated robot only. From that moment the robot and the model predicting its torque were two different robots, with inertias three orders of magnitude apart. The residual $|\tau^{\text{meas}} - \tau^{\text{exp}}|$ became large *always*, with or without contact, and the flag stayed high for over 95% of the swing phase. With the flag stuck at one, terrain search never enters the branch that increases the foot extension, and the command degenerates into a constant.

Table 4: Contact flag versus force measured at the foot, at unchanged threshold. The second row is the diagnosis: an always-high flag is wrong exactly as many times as ours, so ours carried no information.

	misaligned estimator	**aligned**
flag error during stance	7.0%	**0.6**%
error of an *always-high* flag	7.0%	3.2%
does the flag carry information?	**no**	yes
swings without any search reset	20/60	2/60

The fix. A routine rewrites the mass and inertia of every body of the estimator's rigid-body tree with the same corrected values applied to the plant, forces the masked subsystem to re-evaluate and verifies in three successive ways — write, fallback, read-back — stopping with an error if any of them fails. A silent failure here would restore exactly the problem it closes. It is called right after the inertia correction in every task script, in memory, without ever saving the model.

4.3 Re-tuning of the contact threshold

With the aligned estimator the flag works, but at the original threshold [0.04, 0.1, 0.1] it still stays high for 41.9% of the swing, and in 39.7% it is the coxa alone that keeps it high: its p_{90} in swing (0.0421) is just below its p_{10} in stance (0.0469). This has a physical explanation — the coxa rotates about the vertical axis while the contact is vertical, so the contact transmits almost no torque to it, whereas in swing it carries all the lateral acceleration of the leg.

The adopted triple is [0.385, 0.136, 0.105], which places the coxa threshold above its own median in stance, i.e. disables it. **This is a rule change, not a re-tuning:** the flag becomes an OR over two joints. It was decided on the obstacle course, where the outcome is binary and not subject to the variability band, and must be declared as our choice: it is no longer the value present in the reference implementation.

4.4 Slew rate and height saturation

Introducing the feedback controllers required inserting two signal-conditioning blocks on the foot height commands, in order to guarantee the kinematic feasibility of the motion and avoid impulsive transients:

- **Height saturator:** The attitude loop in C3 computes an additive correction Δz_i for each foot, which tends to shift the whole vertical trajectory upwards. Without an upper limit, the maximum height reached by the leg would exceed the robot's admissible workspace, compromising the kinematic reachability of the commanded positions and the stability of the actuators. For this reason, a saturation block was inserted that constrains the maximum foot height within the physical limits of the structure.
- **Rate limiter (slew rate):** The dynamic corrections introduced by C2 (terrain search) and C3 (attitude control) can generate abrupt variations or discontinuities in the height command, resulting in undesired "jerks" of the legs. To attenuate these impulsive stresses, a rate limiter of $\pm 0.4\,\mathrm{m/s}$ was applied. The choice of this value is not arbitrary: since the peak vertical velocity of the nominal gait is $0.375\,\mathrm{m/s}$, the $\pm 0.4\,\mathrm{m/s}$ threshold fully preserves the nominal step trajectory, acting only on the high-frequency peaks produced by the feedback loops.

To avoid methodological asymmetries in the comparisons, these blocks were extended to all three controllers. We formally verified that the performance metrics of C1 are byte-for-byte identical with and without limiters on five of the seven tasks, as well as in the nominal cell of the sixth, and that the same holds for the corresponding variability band on the tasks on which it is measured.

Although this is an implementation detail and not a control architecture, signal conditioning proved decisive: the limiter reduced the lateral drift of C2 on the obstacle course (bringing it from 0.296 to 0.104 m) and suppressed an anomalous sensitivity to measurement variability in the energy computation of C3.

5 Results

All the rows that follow come from a single campaign run after the corrections of Section 4, with the rate limiter present on both plants: seven tasks $\times$ three controllers; in addition, on four tasks, three runs per controller with perturbed z_0. Verdicts in bold pass the $3\times$ rule of Section 3.

5.1 Flat terrain

Table 5: Flat (T2), nominal speed (1.00×). In parentheses the ratio to the variability band.

	C1	C2	C3	C1 → C2	C2 → C3
T2					
max pitch [deg]	0.514	1.95	0.356	**+279%** (16.3)	**−82%** (18.1)
max roll [deg]	0.206	0.280	0.205	+36% (n.c.)	−27% (n.c.)
cost of transport	0.888	0.865	1.11	−3% (n.c.)	**+28%** (27.7)
energy [J]	18.6	18.2	23.3	−2% (n.c.)	**+28%** (28.4)
RMS torque [N m]	0.400	0.403	0.372	+1% (n.c.)	**−8%** (8.0)
peak torque [N m]	1.87	2.02	2.23	+8% (n.c.)	+11% (n.c.)
mean speed [m/s]	0.135	0.136	0.135	**+1%** (6.9)	−0% (n.c.)

On a terrain that is exactly what the trajectory generator believes it to be, terrain search makes body attitude *worse* by a factor of three or four: energy and cost of transport remain unchanged within the variability band on flat ground and improve by two or three percent on the curve. The mechanism keeps working where there is nothing to find, and the body pays for it.

5.2 Differences on T2 as speed varies

A sweep of five speed factors, judged on body bounce with a threshold declared in advance, identifies the envelope of the open-loop gait.

Table 6: Body bounce versus commanded speed factor (T2). Bounce is the peak-to-peak body height within each gait cycle, median over the steady-state cycles. Threshold 10 mm, declared before the sweep.

factor	**C1 [mm]**	**C2 [mm]**	**C3 [mm]**
0.50	2.6	4.3	3.3
1.00	2.9	4.2	3.5
1.20	3.0	3.8	3.8
1.50	3.1	3.1	2.9
2.00	1.2	0.4	2.8

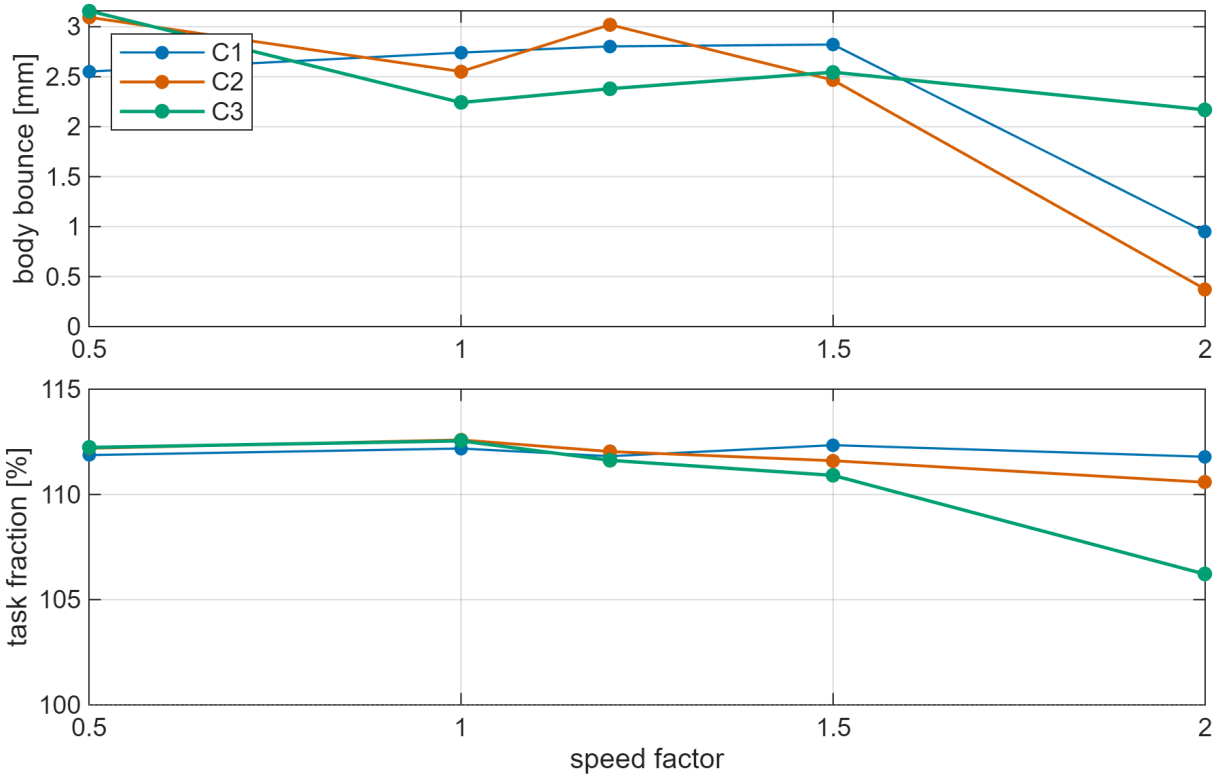


Figure 2: Speed sweep on flat ground (T2) for the three controllers. Top: body bounce, i.e. the maximum deviation of body height from its steady-state median. It stays below 3.2 mm at every speed factor, an order of magnitude under the 10 mm threshold, so the open-loop gait is valid from 0.5× to 2.0×. Bottom: task fraction. C1 and C2 stay between 110% and 113%; only C3 drops, down to 106% at 2.0×. With a 0.5 s period the attitude loop disturbs the foot contact instead of stabilising the body, and the time with fewer than three feet on the ground rises to 29%. As speed increases, the limit is not body stability but the peak torque demanded at the joints.

5.3 Curved trajectory

Table 7: Curved path (T3), nominal speed. In parentheses the ratio to the variability band.

	C1	C2	C3	C1 → C2	C2 → C3
T3					
max pitch [deg]	0.579	1.85	0.392	+**219%** (40.0)	−**79%** (45.9)
max roll [deg]	0.224	0.351	0.210	+**57%** (4.6)	−**40%** (5.1)
cost of transport	0.970	0.951	1.22	−**2%** (3.6)	+**28%** (24.3)
energy [J]	30.3	29.5	37.8	−**3%** (6.4)	+**28%** (22.5)
RMS torque [N m]	0.398	0.389	0.376	−**2%** (7.0)	−**3%** (7.3)
peak torque [N m]	1.91	2.11	2.43	+10% (n.c.)	+15% (n.c.)

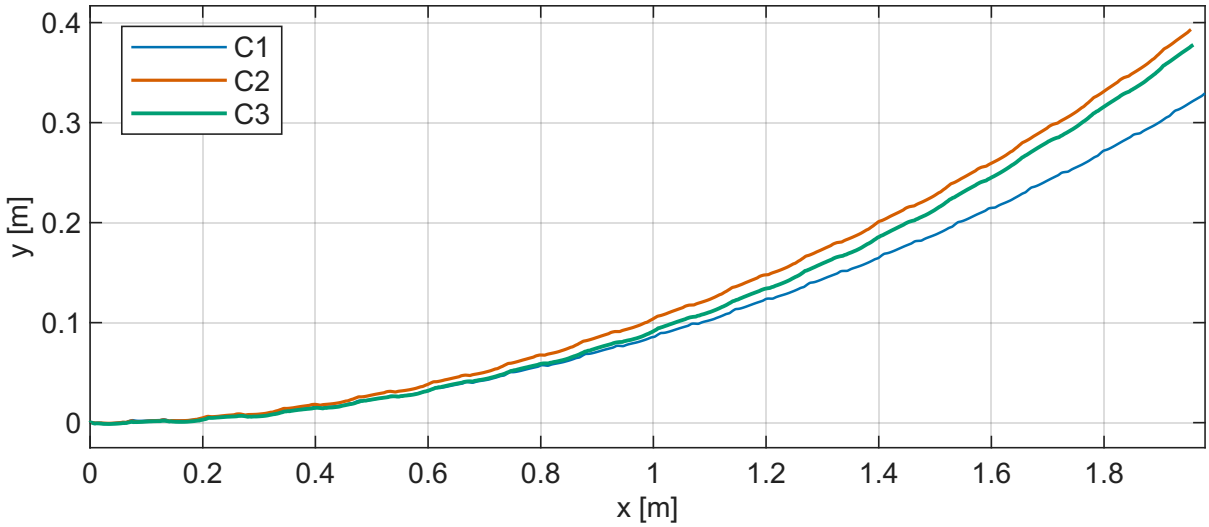


Figure 3: Trajectory of the centre of mass in the horizontal plane with a yaw command of +0.1 rad/s (T3). The three curves overlap almost everywhere: the yaw command is followed in the same way by the three controllers, and the differences in Table 7 on this task concern body attitude, not the path. **The two axes have different scales** — almost two metres on the abscissa versus four decimetres on the ordinate — so the lateral deviation appears much more pronounced than it is.

5.4 Discontinuities: bump and obstacle

Table 8: Bump (T4D) and single obstacle (T5).

	C1	C2	C3	C1 → C2	C2 → C3
T4D					
max pitch [deg]	8.65	10.4	1.98	+**21%** (19.8)	−**81%** (93.8)
max roll [deg]	2.12	1.49	0.611	−**30%** (6.4)	−**59%** (7.6)
cost of transport	1.12	1.04	1.45	−**7%** (5.1)	+**40%** (16.8)
energy [J]	64.1	59.9	83.6	−**7%** (3.5)	+**40%** (18.8)
mean speed [m/s]	0.130	0.133	0.131	+**2%** (3.6)	−**1%** (3.0)
T5					
max pitch [deg]	7.56	6.41	1.73	−**15%** (16.4)	−**73%** (19.8)
max roll [deg]	4.49	2.31	1.13	−49% (n.c.)	−51% (n.c.)
cost of transport	1.20	1.08	1.40	−**10%** (5.2)	+**30%** (17.1)
energy [J]	48.1	44.3	57.5	−**8%** (3.6)	+**30%** (17.2)
RMS torque [N m]	0.408	0.403	0.380	−1% (n.c.)	−**6%** (9.9)
peak torque [N m]	4.46	8.33	4.01	+87% (n.c.)	−52% (n.c.)
obstacle cleared	yes	yes	yes	same	same

Here the picture is reversed, and this is the result that reproduces the claim of the reference work: on a discontinuity the search improves energy and cost of transport on both, and roll on the bump. A solid exception remains on the bump: **peak pitch is** 21% **worse**, at a ratio of 19.8. On a continuous slope the search extends the downhill feet and the body follows the gradient, adapting to the terrain instead of opposing it.

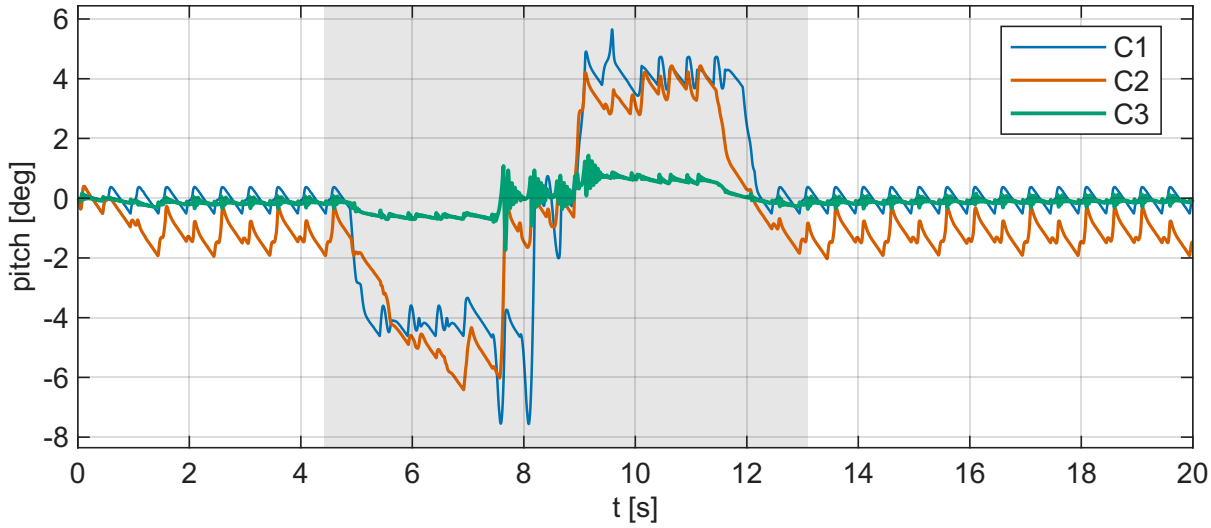


Figure 4: Body pitch while crossing the single obstacle (T5). In grey the window in which at least one foot is on the obstacle, identified kinematically. Outside that window the three curves oscillate around zero with the step amplitude; inside, C1 and C2 make an excursion of several degrees in both directions — first nose up while the front legs climb onto the step, then nose down when the rear ones climb onto it — while C3 stays within about one degree throughout the crossing. It is the same difference that Table 8 reports as −73% peak pitch.

5.5 Ramp

Table 9: 8° ramp (T4). Variability band not measured on this task; no verdicts are issued, but the difference is qualitative and unequivocal.

	C1	C2	C3
T4			
body attitude minus ramp angle [deg]	0.106	0.259	−7.54
peak-to-peak pitch on the ramp [deg]	0.994	1.97	0.839
roll on the ramp [deg]	0.662	0.563	0.314
cost of transport	1.15	1.09	1.58
successful climb	yes	yes	yes

C1 and C2 align the body with the ramp, to within a few tenths of a degree. C3 keeps it at 7.54° from the ramp, i.e. within 0.5° of the *horizontal*, which is exactly what a loop with zero attitude reference is built for, and is also the behaviour that the reference work declares as its own objective. It costs 45% in cost of transport.

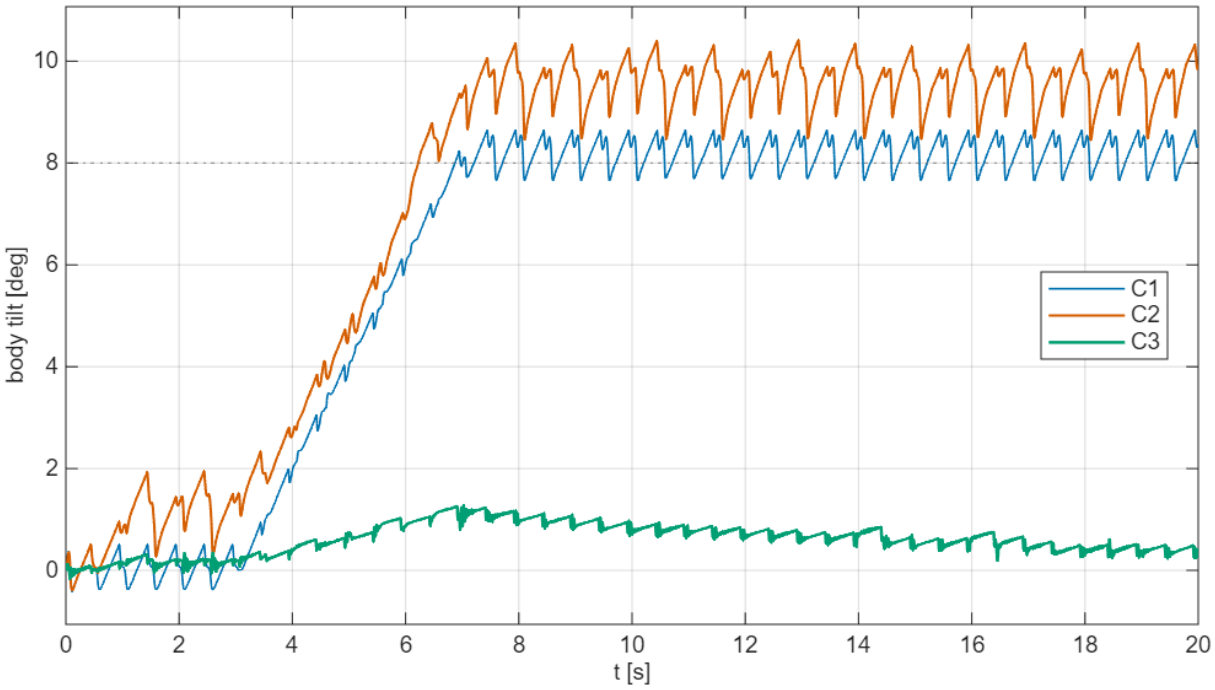


Figure 5: Body inclination while climbing the 8° ramp (T4). The dashed line at 8° is the ramp slope: a curve resting on it describes a trunk *parallel to the ground*. The first 4.5 s are on flat ground; then C1 and C2 rise to the slope and stay there for the rest of the run, while C3 stops below 1.5°, i.e. keeps the trunk horizontal. It is not a smaller inclination of the same thing: they are two different behaviours, which is why the corresponding row of Table 9 carries no percentage. The sawtooth superimposed on all three curves is the step's own oscillation.

5.6 Obstacle course

Table 10: Seven-obstacle course (T6). The run is truncated when a foot reaches the edge of the floor ($x = 3.95$ m), hence the different durations; for C1 the run ends at the maximum duration of 32 s; energy and cost of transport, being time integrals, are excluded for this reason.

	C1	C2	C3
T6			
obstacles crossed (out of 7)	7	7	7
advance [m]	3.65	3.68	3.72
effective duration [s]	32.0	29.3	31.0
max lateral drift [m]	0.067	0.104	0.058
max roll [deg]	9.44	10.0	2.78
max pitch [deg]	14.7	13.6	4.44
peak torque [N m]	7.51	7.97	4.18
rollovers	0	0	0
course declared completed	no	no	yes

The course consists of seven obstacles 35 mm high: four span the whole width and must necessarily be stepped over, three are lateral. The last one ends at 3.388 m, and the floor at 3.95 m: the run is truncated when a foot reaches the edge, so as not to measure a fall from the slab instead of the controller.

All three controllers cross all seven obstacles, without rollovers and without stopping: the advance is 3.65–3.72 m, beyond the 3.61 m position at which the rear leg too has left the last obstacle.

The lateral drift stays, for all three, below the 0.183 m threshold beyond which an obstacle

could have been bypassed rather than stepped over — a threshold derived from the obstacle geometry and from the nominal foot pose, independently of any controller. The three therefore met the same obstacles.

It is on this task that the attitude loop pays off most. C1 and C2 reach 9.4–10.0° of roll and 13.6–14.7° of pitch; C3 stays within 2.8° and 4.4°, i.e. −72% and −67% with respect to C2. Terrain search alone brings no net improvement (+6% of roll and −7% of pitch with respect to C1). Peak torque also improves with C3, which almost halves it: 4.18 N m versus the 7.97 N m of C2. Plausibly, by keeping the trunk horizontal C3 avoids unbalancing the centre of mass and keeps the legs in more favourable kinematic configurations, reducing the lever arms of the weight force and preventing ground impacts from being discharged asymmetrically onto the joints.

Only one controller meets the "course completed" criterion, set before the campaign as remaining a whole gait cycle beyond the last stance on an obstacle. C3 completes its last stance at 29.4 s and has that margin; C1 at 31 s is still on the obstacles and C2 reaches the floor edge first. The distinction is therefore between *crossing* the course, which all three manage, and *completing it with margin*, which only C3 manages.

5.7 Lateral impulse

Table 11: Residual lateral deviation after an impulsive disturbance (T7).

	C1	C2	C3
T7			
residual deviation, 1× impulse [mm]	1.34	6.04	0.164
residual deviation, 16× impulse [mm]	393	400	392
recovers at 16×	no	no	no

At large impulse the three curves coincide within 2%: none of the three recovers a measurable fraction of the deviation. Contact detection is not the mechanism at play — this is an architectural gap common to all three, and the clearest motivation for any future controller on this platform.

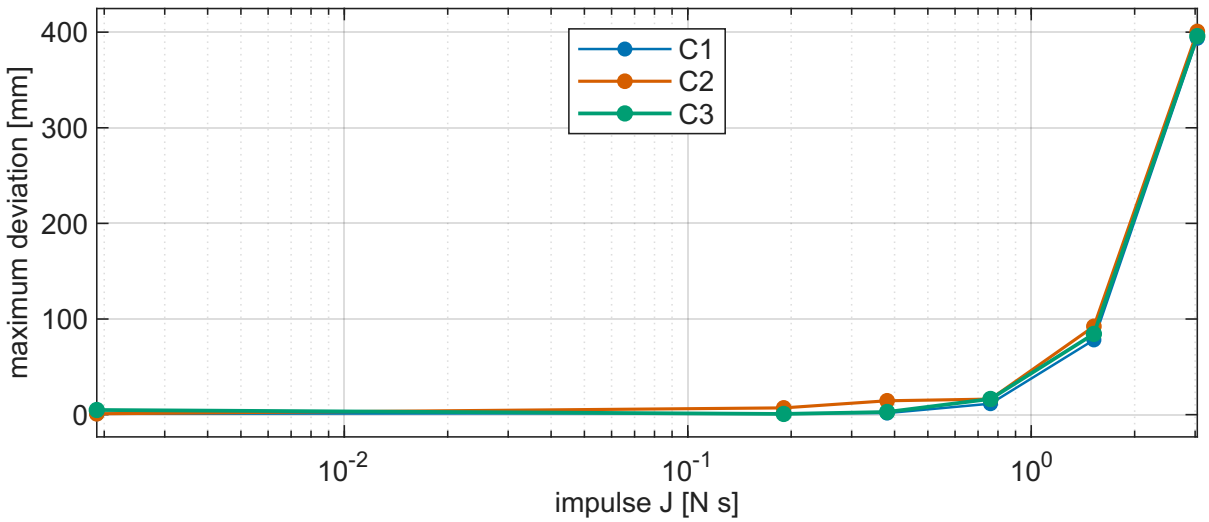


Figure 6: Maximum lateral deviation versus the amplitude of the applied impulse (T7), with a logarithmic abscissa to cover the three decades of the sweep. The three curves are practically indistinguishable over the whole range: up to about 0.7 N s the deviation stays at a few millimetres for all, beyond that it rises sharply to almost 400 mm without any of the three controllers standing out from the others. It is the direct representation of a common architectural gap: neither terrain search nor the attitude loop acts on this disturbance.

6 Feasibility on the real actuators

The campaign runs with ideal actuators: the simulator grants any torque. The AX-12A stalls at 1.5 N m, and the manufacturer recommends working at one fifth of stall for stable motion [2]. Since the joints are position-commanded, the torque is a reaction and not an input, so what follows measures what each controller *demands*, not how it would behave with the real motors.

Table 12: Commanded torque versus the AX-12A limit. RMS is the most loaded joint of each run, as a fraction of stall; "peak" is the maximum torque as a multiple of stall; "over" is the fraction of samples above stall.

	RMS / stall			peak / stall			samples over		
task	C1	C2	C3	C1	C2	C3	C1	C2	C3
T2	46%	44%	39%	1.2	1.3	1.5	0.40%	0.45%	0.38%
T3	46%	42%	39%	1.3	1.4	1.6	0.50%	0.40%	0.43%
T4	44%	44%	45%	1.4	2.3	1.8	0.39%	0.49%	0.48%
T4D	42%	42%	39%	1.7	2.7	2.1	0.45%	0.41%	0.49%
T5	45%	44%	40%	3.0	5.6	2.7	0.61%	0.50%	0.51%
T6	42%	43%	42%	5.0	4.9	2.8	0.91%	0.67%	0.81%
T7	46%	44%	39%	1.2	1.3	1.5	0.41%	0.45%	0.38%

Two regimes, which must be read separately:

- **Walking exceeds the recommended load.** The most loaded joint sits between 39% and 46% of stall on every task and for every controller — 2.3× the one fifth of stall that the manufacturer indicates for stable motion. It stays below stall, but not within the recommended envelope.
- **Impact peaks go beyond stall**, up to 5.0× on the obstacle course, but on at most 0.91% of the samples: they are isolated events, one per foot impact, lasting on the order of tens of milliseconds.

C3 demands the lowest continuous torque on six of the seven tasks (39–45%) and the lowest impact peaks on T5 and T6, which is consistent with an attitude loop that absorbs part of the impact in the body motion rather than in the joints.

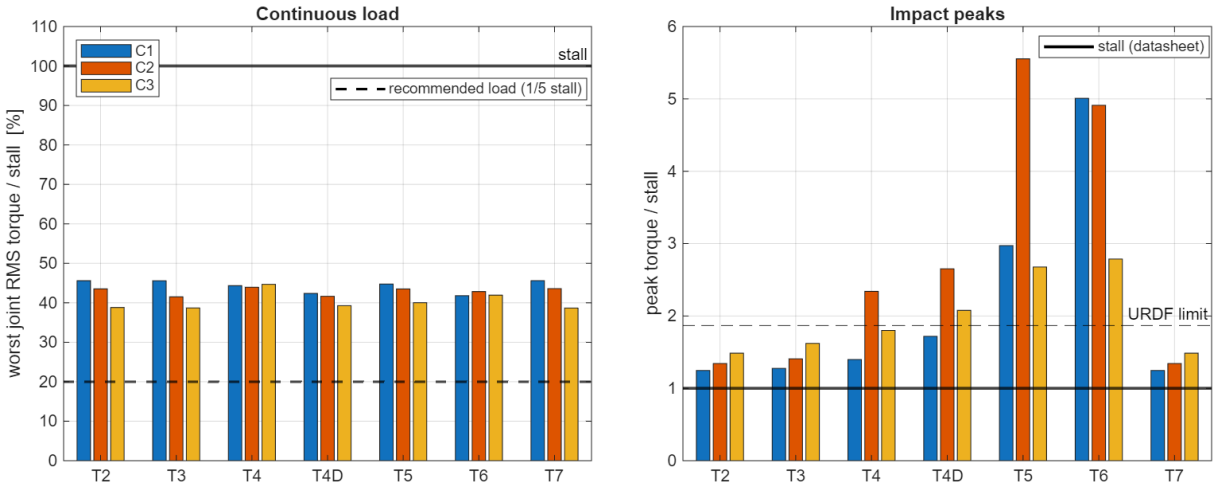


Figure 7: Torque demanded at the joints versus the AX-12A limits, per task and controller. Two distinct regimes must be read: the RMS torque of the most loaded joint, which describes walking and stays steadily below stall but above the continuous load recommended by the manufacturer, and the peak torque, which exceeds stall at stance impacts. The corresponding numerical values are in Table 12.

7 Load transport

All the measurements reported so far evaluate body attitude as a quantity to be minimised, without saying *what keeping it still is for*. The last configuration tested fills this gap: the same plant with an active load, free to move.

7.1 Description of C3P

C3P is C3 with the model's load enabled: a 1 kg cube resting — not constrained — on a 0.1 kg tray rigidly attached to the body. That is 1.1 kg against the 1.585 kg of the unloaded robot: a **69%** increase in mass.

The control law is identical to that of C3: same gains, same saturation, same rate limiter, no re-tuning. The plant is also the same file: the difference between C3 and C3P is the `Commented` state of four blocks (the two solids, the six-degree-of-freedom joint and the contact between cube and tray), switched in memory by a dedicated routine before the simulation. No variant of the model is saved to disk.

> C3P is therefore not a benchmark for C1 and C2: it is a *load case*. It is measured against another configuration with the same load, not against the unloaded controllers.

7.2 Description of the load and measurement criteria

The cube is not welded to the tray: it is connected by a six-degree-of-freedom joint and held up by contact alone, with friction. It can therefore slide, rotate and, in principle, fall.

This choice turns body attitude from a number to be minimised into a *functional requirement*. A pitch of a few degrees is no longer just a worse metric: it is an inclination of the tray that the load sees, and that may be enough to make it walk to the edge. It is the only measurement of the campaign in which the attitude loop has an objective external to the control itself.

The criterion, declared before launching the runs, is twofold:

1. the task must be completed: no rollover, lateral drift within the task's validity threshold, and the path crossed in full;
2. the package must stay on the tray: maximum displacement from the reference below 50 mm, which is the clearance between cube and tray. Below 20 mm we classify it as still.

Asking whether C3 holds the load does not say whether the load needs that controller. Answering the second question requires a comparison with the same load, and we built it without touching the plant: C2P is C3P with the gains of the two attitude PIs set to zero. Same model, same package, same terrain search; the only difference is the loop.

7.3 Results

The tests were run on the three tasks with non-flat terrain — ramp (T4), bump (T4D) and obstacle course (T6) — i.e. those in which the load is actually stressed.

The robot completes the tasks. Without the attitude loop the task does not fail: the distance travelled stays within two percent of that of C3P on all three paths, none of the six runs rolls over, and the RMS torque of the most loaded joint is the same to within a few percent. From the locomotion point of view the two configurations are indistinguishable.

Table 13: Load transport with the attitude loop off (C2P) and on (C3P), on the same three tasks. All quantities are measured within the validity window of the run, which on T4D and T6 ends when a foot reaches the edge of the floor: beyond that instant one would be measuring the fall from the slab. For C2P the package displacement is not reported because the package falls off the tray or exceeds the threshold. The variability band was not measured on these configurations: the differences are observations, without a verdict.

	C2P	C3P
T4 — 8° ramp		
distance travelled [m]	2.39	2.35
max pitch [deg]	12.60	1.33
max roll [deg]	1.00	0.44
RMS torque, worst joint [N m]	0.963	1.004
max package displacement [mm]	—	19
package outcome	**fallen**	stays on the tray
T4D — bump		
distance travelled [m]	3.70	3.71
max pitch [deg]	12.15	1.68
max roll [deg]	1.47	0.45
RMS torque, worst joint [N m]	1.028	0.956
max package displacement [mm]	—	14
package outcome	**at the limit**	stays on the tray
T6 — obstacle course		
distance travelled [m]	3.67	3.72
max pitch [deg]	12.77	3.53
max roll [deg]	11.13	2.07
RMS torque, worst joint [N m]	0.999	0.989
max package displacement [mm]	—	20
package outcome	**at the limit**	stays on the tray

The load is not stable. On all three tasks the package of C2P exceeds the 50 mm clearance between cube and tray, and does so early: between 6.4 and 8.5 s, i.e. within the first nine seconds of each run. The way it exceeds it, however, is not the same. On the ramp the travel of the cube is monotonic: it crosses the threshold and never comes back, until it leaves the robot and falls out of the scene. On the bump and on the obstacle course the cube crosses the threshold several times — eight and five times respectively — banging up to about 92 mm and coming back, and ends the valid window at 31 mm on T4D and at 91 mm on T6. In none of the three cases is the load delivered, but only on the ramp can one speak of a clean fall. With the loop on, the maximum displacement stays at 19 mm on T4, 14 mm on T4D and 20 mm on T6: the threshold is never reached, and the vertical component stays below 6 mm, i.e. the cube does not bounce. Applying the threshold declared in advance, C3P is *still* on T4 and T4D and just above 20 mm on T6, where it must be classified as "slides but stays on the tray".

Benefits given by attitude control. Maximum pitch goes from about 12–13° to 1.3–3.5°, a factor between four and ten depending on the task; on the obstacle course roll also drops from 11.1° to 2.1°. Twelve degrees is the inclination that the tray presents to the cube, and it is enough to make it walk to the edge.

> This is the measurement that the previous sections lacked. So far body attitude has been treated as a quantity to be minimised, without anything saying how small it had to be: here it has a threshold external to the control, and exceeding it means losing the load. The attitude loop does not improve transport, it makes it possible.

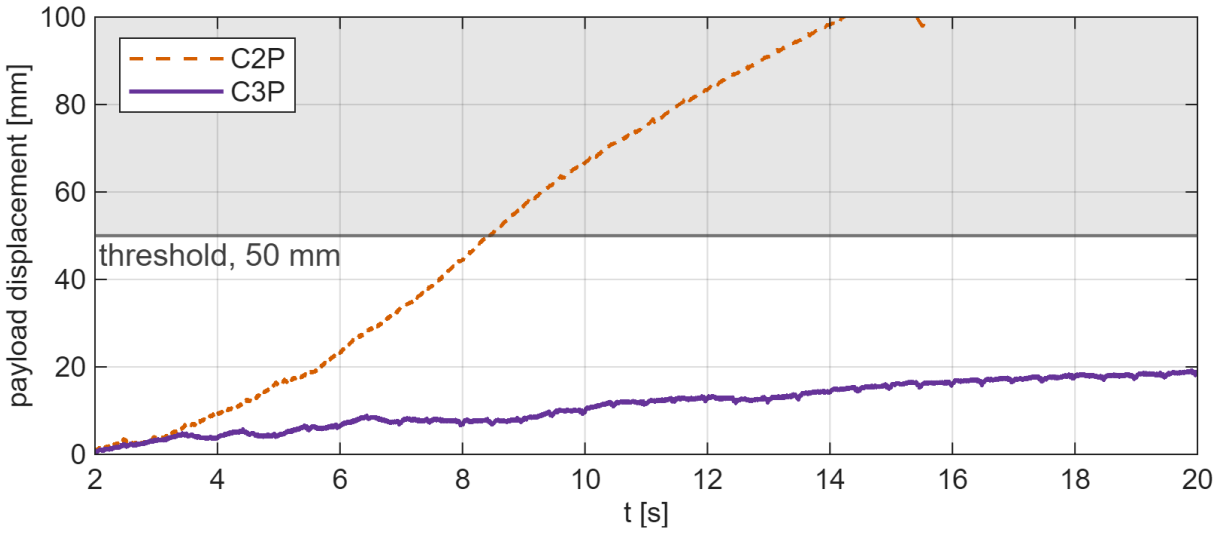


Figure 8: Displacement of the package from the reference on the 8° ramp (T4), with the attitude loop off (C2P) and on (C3P). The line and the grey band mark the 50 mm clearance between cube and tray: above it, the cube is no longer contained by the tray. C2P exceeds it after 8.4 s and continues without coming back — beyond the frame the curve reaches tens of metres, because the cube leaves the robot and falls. C3P stays below 20 mm for the whole run. The measurement starts at 2 s, after the settling transient on which the reference is defined.

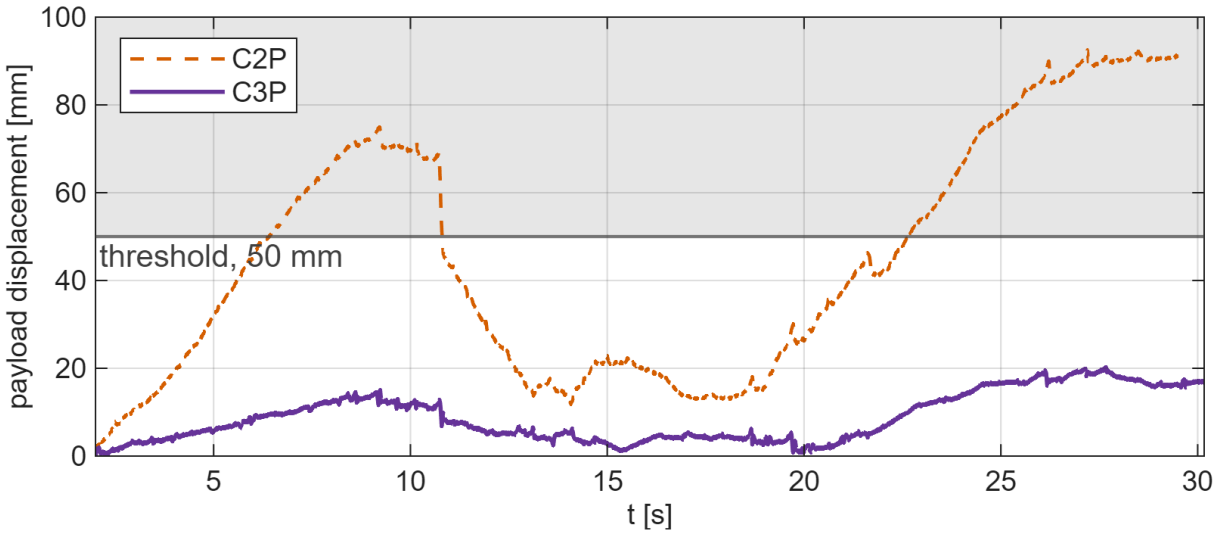


Figure 9: The same measurement on the obstacle course (T6). Here the behaviour of C2P differs from that on the ramp: the cube exceeds the clearance as early as 6.4 s, but then **comes back in and goes out again several times** — five threshold crossings — until it ends the valid window at 91 mm. It is therefore not a clean fall as on T4, but a load banging back and forth well beyond the limits of the tray; the criterion nevertheless classifies it as not delivered. C3P never reaches the threshold. Both curves stop when a foot reaches the edge of the floor (29.5 s for C2P, 30.2 s for C3P): beyond that instant the displacement would measure the fall from the slab.

7.4 Joint torque

With the package on board, the RMS torque of the most loaded joint is about 1.0 N m on all three tasks, i.e. 64–69% **of the stall torque** of the AX-12A (1.5 N m) and 3.3 times the continuous load that the manufacturer recommends for stable motion, one fifth of stall. Impact spikes reach 2.3–3.1 times stall.

This price is due to the load, not to the loop: C2P and C3P demand the same torque to within a few percent. Switching off the attitude loop does not relieve the joints, it only makes the package fall.

> Load transport is therefore the case in which the gap between simulation and hardware weighs the most. In the campaign the actuators are ideal and torque is a reaction, not an input: what we measure is what the controller *demands*. On real servos this is the configuration that would be the first to stop working.

8 Stability analysis

The alternating tripod is built on a *static* stability assumption: at every instant three feet are on the ground and the projection of the centre of mass must fall inside the triangle they form. We wanted to verify it rather than assume it, instrumenting the plant with the online computation of the centre of pressure and of the margin with respect to the support polygon.

8.1 ZMP computation

The Froude number of the nominal gait is

$$\mathrm{Fr} = \frac{v^2}{g\,h} = \frac{0.12^2}{9.81 \times 0.154} = 0.0095, \tag{7}$$

i.e. the inertial forces amount to one percent of gravity. In this regime the ZMP collapses onto the ground projection of the centre of mass within the precision of the measurement, and the

dynamic criterion degenerates into the static one on which the gait is built. We therefore do not expect this measurement to discriminate between the controllers: we expect it to *confirm or refute the assumption* on which everything else rests.

From the contact forces measured at the six feet we compute the centre of pressure on the reference plane $z = 0$:

$$x_{\mathrm{ZMP}} = \frac{\sum_i (x_i F_{z,i} - z_i F_{x,i})}{\sum_i F_{z,i}}, \qquad y_{\mathrm{ZMP}} = \frac{\sum_i (y_i F_{z,i} - z_i F_{y,i})}{\sum_i F_{z,i}}, \tag{8}$$

where the sum runs over the stance legs only, i.e. those whose vertical force exceeds the threshold $F_{\min} = 2\,\mathrm{N}$, about 39 % of the static load of one foot (5.18 N out of a total weight of 15.6 N). The *margin* is the signed distance from the nearest side of the convex polygon formed by these legs, positive inside. At least three legs above threshold are needed for the polygon to be defined: with fewer than three the margin is not computed. The computation block is present only in the model without the attitude loop, so the measurement concerns C1 and C2; we report C2.

8.2 Flat terrain

Figure 10 shows the margin of C2 in straight walking on flat ground (T2) at nominal speed, for 10 s, i.e. ten gait cycles. The tripod switches every 0.5 s.

The sign. At every instant at which it can be computed, the margin is positive: minimum 0.018 m, no negative sample. At steady state ($t \geq 2\,\mathrm{s}$) the median is 0.115 m, and half of the samples fall between 0.102 m and 0.123 m.

Comparison with geometry. With the nominal foot positions, the tripod vertices are 0.224 m and 0.243 m from the body axis, and the distance from the body centre to the nearest side of the support triangle is 0.118 m (dashed line in the figure). The measurement settles on this value: the steady-state median margin is 0.115 m, and at mid-stance, when the feet pass through the nominal position, it rises to 0.127 m.

The structure. The pattern repeats identically at every tripod switch and is arch-shaped: the margin grows from the start of stance, peaks at mid-stance, when the support triangle is centred on the body, and decreases towards the end, when the feet have slid by 3 cm with respect to the body and the triangle has shifted. The deepest dips, down to 0.02–0.05 m, occur close to the tripod switch, during the transfer of the load from one tripod to the other.

High-frequency noise in the margin. As can be seen from the time profile, the stability margin shows an evident high-frequency noise component. This fluctuation does not represent a real instability of the motion, but rather an artefact arising from the computation of the contact forces: both the ZMP coordinates (obtained as the average of the contact positions weighted by the normal forces) and the definition of the vertices of the active support polygon depend directly on the signals returned by the *Spatial Contact Force* blocks. These force sensors generate high-frequency micro-oscillations during the impact and sliding phases of the feet on the ground, which propagate directly to the margin estimate.

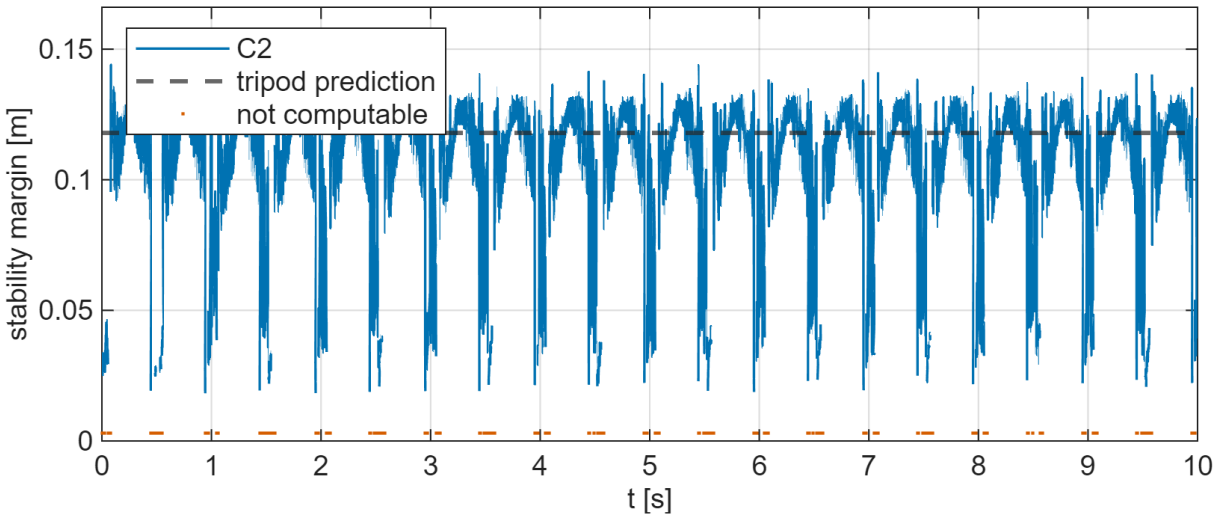


Figure 10: Stability margin of controller C2 in straight walking on flat ground (T2) at nominal speed (0.12 m/s, period 1 s), for ten cycles. The margin is the signed distance of the ZMP from the nearest side of the polygon formed by the stance feet: positive if the ZMP is inside the polygon.

8.3 Non-computable instants

The margin cannot be computed for 10 % of the time, over 160 intervals: most of them last a single sample (1 ms), the longest 39 ms. All of them fall within 0.1 s of the tripod switch. With a duty factor of 0.5 one tripod lifts off at the same instant in which the other lands, with no phase with six feet on the ground: while the load passes from one tripod to the other, some feet oscillate around the 2 N threshold, entering and leaving the count, and for brief instants fewer than three feet are above threshold.

The robot does not lose support: a foot with force below threshold still carries load, and the robot advances without rollovers and with limited attitude. The non-computable instants are therefore a limit of the measurement, not of the robot. The threshold is a compromise: lowering it would reduce the intervals, but barely touching feet would become vertices of the polygon, and the margin would be overestimated.

> On flat ground the measurement reproduces the geometric prediction, and at every instant at which it is defined (90 % of the time) the centre of pressure is inside the support polygon. The static stability assumption on which the tripod is built is therefore verified.

9 Limitations and future work

- Ideal actuators. The joints are position-commanded and the simulator provides whatever torque the command requires. Section 6 therefore measures the *demand*, not the achievable behaviour. A campaign with the AX-12A speed and torque limits in the loop would probably compress the differences, because C3 is the controller that demands the largest additional foot travel per unit time.
- No hardware. Nothing in this report was measured on a physical PhantomX. The platform, the URDF and the servo datasheet are real.
- Static attitude lever arms. The coordinates (x, y) used to compute the height correction $\Delta z_i = x u_\theta + y u_\phi$ are fixed at the mean values identified during walking. Although this simplification proved adequate for the tasks analysed, real-time estimation of the instantaneous foot positions $(x(t), y(t))$ would allow a more accurate distribution of the attitude corrections in more complex operating scenarios.

10 Contributions by team member

Eros Cribello Implementation of the terrain-search layer following the reference paper (C2), including the inverse-dynamics residual and the per-leg contact flag; design, implementation and empirical tuning of the attitude loop (C3) — first pitch, then roll — and of the saturation and rate limiter that bound its authority; quasi-static stability check, with ZMP computation.

Andrea Russo Construction of the Simscape Multibody plant from the PhantomX URDF; implementation of the open-loop tripod generator (C1); design of the scripts of the seven tasks; the feasibility analysis against the servo datasheet.

Both Choice of the three controllers; the entire measurement campaign and its automation (controller selection in a single place, batch execution, conversion of the results, generation of the comparison table); correction of the two bench flaws of Section 4; reconstruction and justification of the gait parameters (Section 2); load tests on C2P and C3P; interpretation of the results.

11 Conclusions

Three controllers for a PhantomX hexapod were compared on a single plant and on seven tasks, with a protocol that declares a difference only when it exceeds three times the measured variability band.

- The terrain search of [1], taken alone, has a higher cost on known terrain: pitch worsens by 219–279% with no energy return. On irregular terrain it improves roll (−30% on the bump), cost of transport (−7 to −10%) and speed (+2% on the bump), while worsening pitch on the bump.
- The C3 loop improves the robot's attitude on every task: pitch from −73 to −82% at a ratio of 18 to 94 times the variability band and, on the 8° ramp, a qualitatively different behaviour (trunk held horizontal instead of parallel to the slope). However, it consumes 28–40% more energy than the previous case.
- On real servos, walking alone would run at 39–46% of stall on the most loaded joint — 2.3× the continuous load recommended by the manufacturer — with impact peaks on about 0.91% of the samples.

References

[1] S. Arrigoni, M. Zangrandi, G. Bianchi, F. Braghin, "Control of a Hexapod Robot Considering Terrain Interaction," *Robotics*, vol. 13, no. 10, art. 142, 2024.

[2] ROBOTIS, *Dynamixel AX-12A e-Manual*, ROBOTIS Co., Ltd. `https://emanual.robotis.com/docs/en/dxl/ax/ax-12a/`

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